

P-Stratum: a foundation for higher layer anonymity systems

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ABSTRACT

This document defines the "Proximate stratum", a network layer built upon the heterogeneous technologies of the ISO-OSI layer 2 ("Data Link Layer"), whose purpose is to enumerate neighbor nodes and to establish an encrypted, reliable, flow-controlled, full duplex connection with a neighbor while solving all issues related to packet fragmentation. Each pair of neighboring P-Stratum nodes agree upon a constant transmission rate upon link setup, using chaffs to shape traffic, a required feature to thwart traffic analysis. It also provides functionalities to anonymously detect shared peers, a necessary trait to avoid cycles during topology discovery at the upper layer.

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1 INTRODUCTION

The TCP-IP protocols were born in a world that was entirely different from the one in which they currently operate.

Fundamentally, the Internet's building blocks are the same core protocols adopted by the ARPANET more than four decades ago, above which new ones were built to extend functionalities matching new requisites (i.e., IPsec, DNS, ...) or to fix lacks in the design (i.e., BGP, ...).

2 GOALS AND PURPOSE

P-Stratum is designed to communicate with a one hop distant nodes that in this stratum context we designate exclusively with the term **peer**. Its objective is to create a uniform graph as characteristics and topology for the upper layer, starting from heterogeneous underlying technologies.

The P-Stratum does mainly five things:

1. discovers the active physical links and their relative characteristics with which the node can connect to peers,
2. announces itself on these physical links and waits for the peer's response,
3. creates an **association** by performing an handshake between two peers, i.e. an exchange of DH keys which agrees on a shared secret (see section 2.5), and
4. manages a two way **connection** that marshal data between two peers encrypting with the shared secret symmetric cryptography, (see section 2.7)
5. notifies the peer about the obfuscated degree of the node, detecting and not counting mutual peers that leads to d=2 cycles at upper level. (see section 2.8)

2.1 Document Structure

2.2 Terms and Definitions

The keywords "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 namely RFC2119[1] and RFC8174[2] when, and only when, they appear in all capitals, as shown here.

Commonly used terms in the document are described below.

Nexus Fundamenta denotes the heterogeneous technologies of the ISO-OSI layer 2 ("Data Link Layer") commonly implemented in the network card driver.

Stratum Data Unit (SDU) is information that is effectively transmitted as a single unit among peer entities of at a lower bound of a stratum.

Process Logical Unit (PLU) is how the information is yielded by a process inside a stratum to be aggregated or split in SDUs.

Data is :

- **submitted** when P-Stratum get it by an overlying stratum to be processed,
- **sent** when P-Stratum submit processed data to an underlying stratum,
- **received** when P-Stratum get it from an underlying one, and
- **delivered** when, after being processed by P-Stratum, are returned to the overlying stratum.

2.3 Notational Conventions

Message Sequence Charts (MSC) format is described in ITU Z.120[3] recommendation.

Packet diagrams in this document use the format described in Section 3.1 of RFC2360[4], with the following additional conventions:

We can then identify two sets of activities conducted by the P-Stratum, **Control Plane** ones necessary in order to enable the transfer of data submitted to a peer, like discovering peers

and establishing, managing and tearing down associations and connections and **Data Plane** ones that conveys data submitted by the upper stratum to a peer.

The stratum design is a kind of generalized MACsec[5] (IEEE 802.1AE) as it can mitigate attacks by defining a secure channel between peers to provide data confidentiality, data integrity, and data origin authentication. Unlike MACsec, the secure channel operates on any Nexus Fundamenta; the trade-off is that it can not take advantage of the dedicated Layer 2 features (for example in the switches). Like TARANET[6] "Link Encryption and Padding", it prevents an adversary observing a data flow between two honest peers from distinguishing different type of submitted data including overlying stratum chaff (used to prevent traffic analysis) from actual data.

The stack is built upon the essential functions of the underlying networks usually implemented in hardware and network driver of the device itself. Its construction (see figure 1) begins with a series of specific software modules called **adapters** for each underlying technology. On the contrary, the P-Stratum **core** contains functionalities that are not dependent on the underlying link technologies like fragmentation engine, encrypted connections handshaking, peer table maintaining, sending and receiving queue processing.

Figure 1. P-Stratum design

Packet is the name used uniquely for the SDU of P-Stratum inside the Nova stack. When some data is submitted to the P-Stratum, it breaks the data into a **sequence** of packets, that is the SLU for all P-Stratum processes. If data to be transmitted doesn't exceeds the MTU size of the adapter, P-Stratum produce a single packet sequence.

Control Plane packets are UDP-like, connectionless; a sequence is valid if all the necessary packets are received, namely to have received all required fields of the announced sequence type.

Data Plane instead is designed as a timer-based encrypted transport protocol between two peers to provide hazard-free connection management without explicit packet exchanges. It implements a sliding window flow control mechanism like one in the Delta-t[7] protocol extended with a rate control to avoid congestion.

To understand how the P-Stratum handles the transmission queue we must take into account the following constraints.

The overlying R-Stratum requires that P-Stratum provides encryption of the data exchanged between two peers. Also, R-Stratum to route own SDU needs it as a whole, but by having its header, it can do some job in advance to optimize the management of flows. However the P-Stratum must be agnostic to the data structures contained in data submitted by R-Stratum (like SLA and payments), vice versa the latter must not do any consideration concerning underlying logic such as the fragmentation of the data.

It follows that when the upper stratum submit data, it must also provide how many bytes from the beginning of the submission are priorities, i.e., it is desirable that packets of the sequence that contains them should be transmitted one after the other before passing other sequences and must provide a kind of priority between subsequent submissions. Further, it follows that P-Stratum must be able of individually decrypt a received packet and this implies that, in the transmission process, the encryption engine must operate immediately after the fragmentation management engine and that the latter must take into account additional bytes required in the packet by the first to comply with Nexus MTU.

Although the P-Stratum knows nothing about the negotiated service level agreement along the R-Stratum route, the first exposes:

- data structures that contain bandwidth, jitter and latency of a link so that SLAs are negotiated compatible with the characteristics exposed by Nexus Fundamenta,

- methods to submit data and to manage its throttling on the physical network,
- events that notify that a delivery is available.

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95 2.4 Adapters

An **adapter** is the way from the core of the P-Stratum through any lower-layer protocols to the physical media connecting to another system (for sending and receiving data on a physical medium such as Ethernet, Wifi, MPLS, Bluetooth, etc...). The same adapter can be instantiated many time to manage, for example, multiple network cards.

100 The process to assign a node to an adapter is called **binding**. A many to many (M: N) relationships can be established between the two kind of entities.

From an implementation perspective, the adapter must comply with a programming interface (core to adapters interface **C2AI**) that exposes methods required by the core of P-Stratum. Few examples are:

- 105 • the discovery of the maximum transmission unit (MTU) that it's used to handle fragmentation,
- the setup of MPL, R, and A protocol parameters, related to physical network speed (see section 2.7),
- the detection of the congestion to manage correctly the throttling and buffering of data to be transmitted, and
- 110 • the implementation of the discovery of peers that depend on the availability of a broadcast mechanism on the underlying network.

Nova requires that every nexus fundamenta provide an MTU of 1280 bytes or higher, by leveraging the similar choice done in IPv6[8]. On any link that cannot convey a 1280-byte 115 packet in one piece, link-specific fragmentation and reassembly must be provided at a nexus fundamenta below P-Stratum.

Special Adapters

2.4.1 Internal virtual adapter

120 IVP is used internally in a device to connect endpoint applications and the internal device router. The aim to use this virtual adapter is to enhance privacy as the originator traffic become indistinguishable from rerouted traffic outside the device. In some way, it resembles a traditional Loopback interface.

2.4.2 NOVA over TCP-IP virtual adapter

125 It is used to route NOVA traffic over TCP-IP, it works like an overlay network access point. This feature is necessary to bootstrap the network at the beginning of its deployment.

2.5 Discovery and Association process

When a node joins the network, P-Stratum performs the mapping of the underlying physical network topology in a neighborhood table, finding peers and associating their link-layer address and Nova addresses. It resemble the functionality done by the Address Resolution Protocol 130 (ARP)[9] protocol in IPv4 or Neighbor Discovery Protocol (NDP)[10] in IPv6.

Initially, a peer doesn't know long-term public keys of the other party to establish an encrypted communication, nor can send it's own in the clear over the network, nor they have any shared secret to sending encrypted transport message. The peers, to get to know each other both of this information, perform a Diffie-Hellman key exchange using a NOISE interactive 135 pattern (see section 3.1 section in the Appendix for noise nomenclature).

Noise interactive pattern		D&A Sequences exchange
XX(s, rs):		
		JP --(JOIN)--> RP
-> e		RP --(JOIN-REPLY)--> JP
140 <- e, ee, s, es		JP --(HANDSHAKE)--> RP
-> s, se		RP --(HANDSHAKE-REPLY)--> JP

The mapping should be authenticated, and this authentication should be computational more expensive for the joining peer (JP) because node key pairs are locally long-term generated and an attacker can perform a DOS by continuously creating identities.

145 To achieve this outcome, the JOIN-REPLY sequence contains a difficulty field expressed in Bitcoin format[11]. The hash of the ephemeral public key of JP, a key that MUST be sent to the receiving peer (RP) in the HANDSHAKE sequence, MUST be less than difficulty.

2.6 Paths

150 A **path** is an established P-Stratum association between two peers. It is a 2-tuple (Peer address, Adapter) representing the bi-directional way that the traffic has to use to move from a node to a peer.

When a peer is reachable through two different adapters (for example wifi and wire), then there is a **multi-path** condition that MUST be managed accordingly to the multi-path policy of the node.

155 Load Balancing and Failover configurations, in which a single P-Stratum adapter represent for example multiple network cards on the same underlying network, should be managed by underlying network device driver.

2.7 Connection process

160 The design objectives of P-Stratum data plane were to provide minimum packet exchange for request/response transactions, and high throughput for bulk data transport, while assuring secure communications and submitted message boundary preservation. It supports sliding window flow control, error control of lost, damaged, duplicated, and out-of-sequence packets. It uses the sequence header fields to provide data plane transmission assurance through a positive acknowledgment/retransmission mechanism, and bounds on packet lifetime. It also provides a
165 negative acknowledgment mechanism (DATA-NACK) as efficiency and diagnostic aid.

An important feature of the data plane is its timer based connection management strategy which does not exchange packets to open or close a connection. A connection in P-Stratum terminology means the time during which state information is maintained at each communicating end.

170 Connection opening stands for reliably establishing this state and connection closing stands for reliably and unambiguously deallocating this state. When a connection is in a default state, no state need to be retained.

175 The timer-based connection management mechanism automatically recognizes when a connection is or is not in the default state and allocates and deallocates connection state without connection management packet exchanges. During a connection, the sender and receiver maintain a send-timer and a receive-timer in addition to maintaining state.

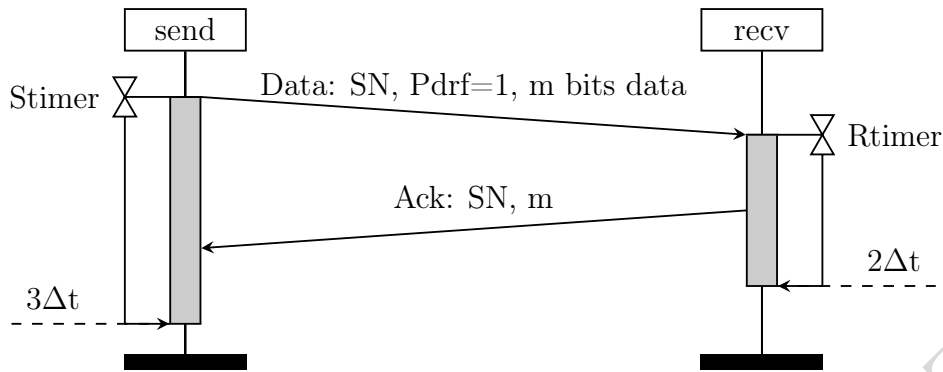
The receive-timer rules assure that state is maintained long enough so that all old duplicates are detectable, assuring reliable connection opening, and that the node will not accept any packets for a safe period on crash recovery.

180 The send-timer rules assure that a PSeqCnts (the counter of the packet in a sequence) value will not be reused until all associated packets and DATA-ACK expire even if there is a node crash.

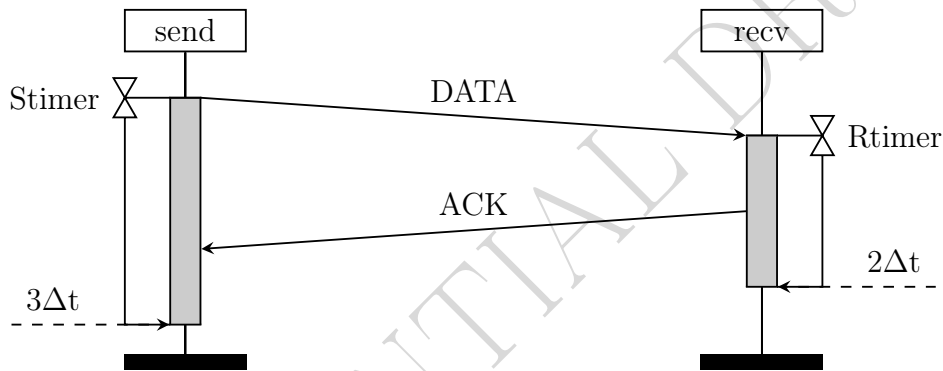
185 The two timers together assure that the state is maintained long enough to guarantee a graceful close and that expired PSeqCnt are generated and accepted. It defines and justifies the receive-timer timeout value as $2\Delta t$, and the send-timer timeout value as $3\Delta t$ where Δt is given as follows

$$\Delta t = MPL + R + A \quad (1)$$

MSC 1 Single Data and Ack Exchange



MSC 2 Mono-directional Multiple Data and Ack Exchange



where:

MPL stands for the maximum packet lifetime or the longest time a packet can exist within the network,

190 R stands for the maximum time a sender will keep retransmitting a packet,

A stands for the maximum time a receiver will wait before sending an DATA-ACK

T stands for the maximum new PSeqCnts generation rate or maximum transmission rate.

For reliable and correct operation, the protocol assumes bounds on the MPL, R, and A values, these bounds are dependent on the physical network over which data is transported.

195 2.8 Degree notification process

that is:

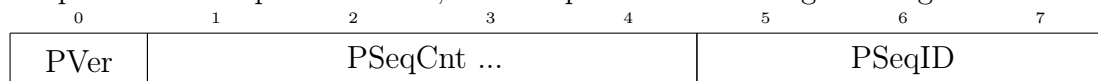
- if a peer have no other edge other than one connecting to the node or their common peers then $\langle s^{i+1} \rangle = 0$,
- else $\langle s^{i+1} \rangle = |s^{i+1}| + \text{noise}$ - common peers

200 that is the degree of next node plus a positive random noise to hide the network topology. The amount of the noise to be added in order to appropriately hide a Nova topology will be investigated in future efforts.

2.9 Packets and Sequences

2.9.1 Packet header

All packets have a packet header, used for protocol versioning and fragmentation management.



This header contains 3 fields:

PVer (1 byte) Each sent packet starts with a single leading byte, which indicates the version of the P-Stratum protocol used. A version of 0 means that no change is necessary, while a non-zero version suggests that the peer has deviated from the protocol specified initially within this document. A node MUST reject handshake attempts initiated with an unknown version.

PSeqCnt (4 bytes) The Sequence Counter is used to enumerate the fragment so it can be reassembled in the correct order on the peer. When PSeqCnts overflow, it wraps around, avoiding 00000000 that is reserved exclusively for the first packet of a sequence. It is important to avoid two packets or their DATA-ACK with the same PSeqCnt existing in the network at the same instant. The protocol requires the following inequality to hold to satisfy the unique PSeqCnt condition:

$$2^n > (2MPL + R + A)T$$

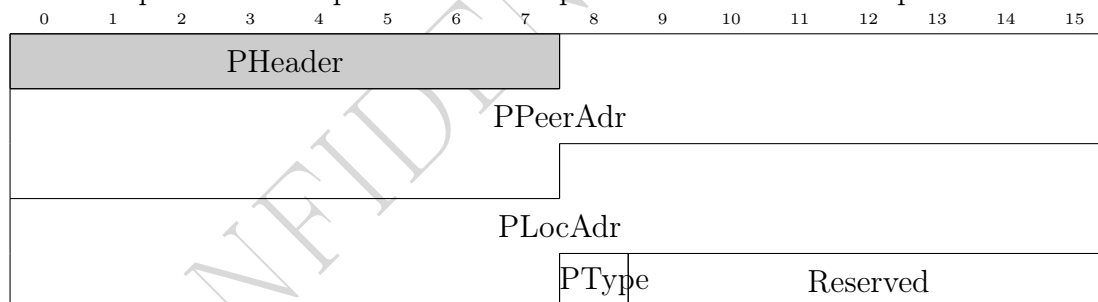
In other words, $1/T$ indicates the time spanned between two PSeqCnts generations.

PSeqID (3 bytes) The second field is Sequence ID a sequential number representing a flow of packets that conveys to the peer a unit of information (i.e. an R-Stratum SDU)

Both these fields are also used to manage out-of-order packets decryption (NOISE 11.4. Out-of-order transport messages).

2.9.2 Sequence header

The first packet of a sequence has a "sequence header" after the packet header.



This header contains 3 fields:

PPeerAdr (32 bytes) The address of remote peer, the destination of the packet, all 0 if it is a broadcast.

PLocAdr (32 bytes) The address of local node, the originator of the packet.

PSeqT (1 bytes) Sequence Type is set accordingly to the kind of packet according to the table 1.

Data Plane packet has a lower priority in the sending queue than Control Plane packets to ensure that all P-Stratum o control operations work as expected.

2.9.3 Packet trailer

All packets have a packet trailer, that contain:

PCRC32C (8 bytes) It is the a CRC32C checksum of all data contained in the packet. If the checksum verification fail a DATA-NACK is emitted with "checksum failed" flag set.

Type	Description	Priority	See section
0	DATA	0	2.9.9
1	DATA-ACK	1	2.9.11
2	DATA-NACK	1	2.9.12
3	DATA-RENDEZVOUS	1	2.9.13
32	JOIN	2	2.9.4
33	JOIN-REPLY	2	2.9.5
34	HANDSHAKE	2	2.9.6
35	HANDSHAKE-REPLY	2	2.9.7
36	LEAVE	2	2.9.8

Table 1. Sequence Types

240 2.9.4 JOIN Sequence

As soon as an adapter is activated, each node bond to that adapter broadcast a Join sequence. Join payload fields set contain:

PNFAdr (6 bytes for ethernet) It is the Nexus Fundamenta address of local peer (MAC Address).

245 **PSignature** (64 bytes) It is the digital signature of PHeader+SHeader+PNFAdr done with node long-term key-pair. It create a computational expense to join the network and is used at a later stage to authenticate the MAC address provided.

Receiving peer (RP) create a entry in the peers table, associated to the joining peer (SP) tagged with "Reachable" state. RP start as soon as possible an handshake to exchange long
250 term public keys and a shared secret to encrypt with efficient symmetric encryption the data packets.

Join Sequence is used also to change nexus fundamenta address of an already joined node after digital signature verification.

2.9.5 JOIN-REPLY Sequence

255 As soon as a receiving peer receive a JOIN sequence broadcast, it announces self by sending a JOIN-REPLY containing the following payload:

PEphPubKeySender (32 bytes) The compressed public key of an ephemeral key-pair created by the receiver peer.

PDifficulty (4 bytes) A difficulty expressed in Bitcoin format[11] required to comply.

260 2.9.6 HANDSHAKE Sequence

2.9.7 HANDSHAKE-REPLY Sequence

As soon as the handshake is completed the peers table entry is updated in "Connected" state.

2.9.8 LEAVE Sequence

265 Before a deactivation of an adapter, each node bond to that adapter broadcast a sequence containing these fields:

:

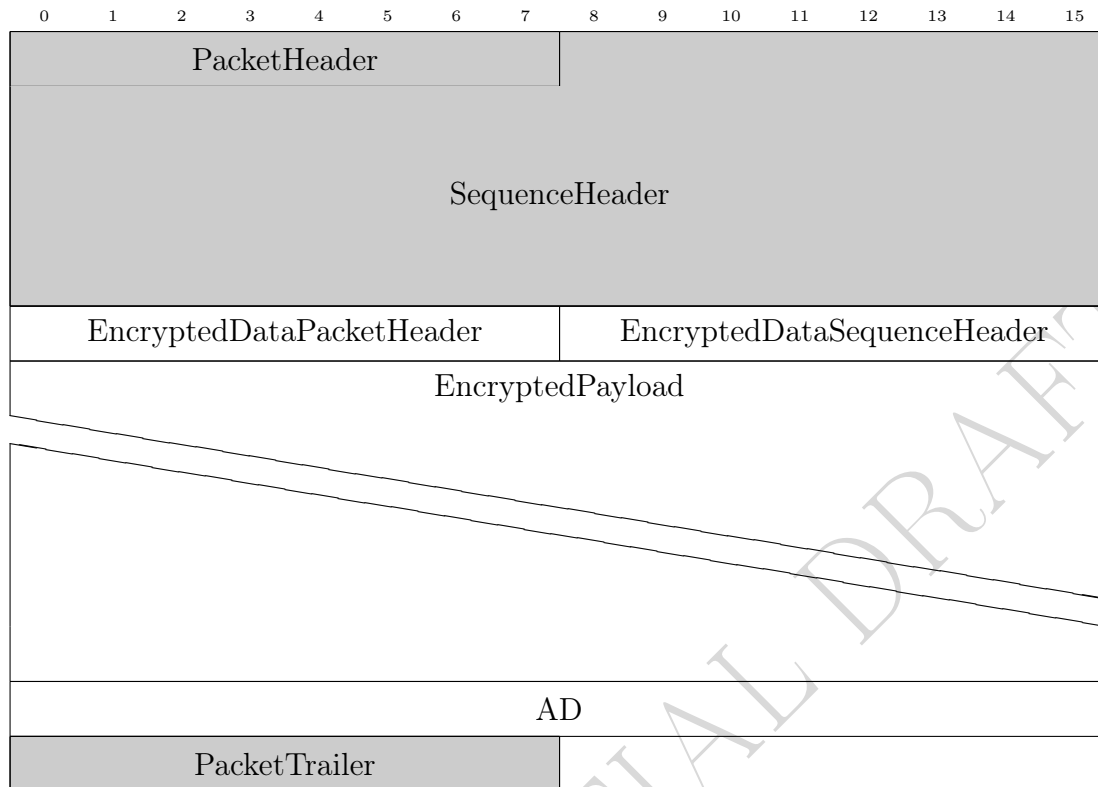
PSignature (64 bytes) It is the digital signature of PHeader+SHeader done with node long-term key-pair used to authenticate the leave.

270 Receiving hosts stop accepting packets and remove the peer entry from the peers table as soon as the signature is verified.

PING-PONG ? If no packets are exchanged between the peers, and it stays in the "Connected" state for longer than its "Connection Time" value, the entry changes to the "Stale" state.

2.9.9 DATA Sequence

275 When P-Stratum receive from R-Stratum the flowlet to be transmitted, it have to be broken into parts to manage the required fragmentation.



where:

PacketHeader (8 byte) is defined in section 2.9.1

280 **SequenceHeader** (70 byte) is defined in section 2.9.2. Only in the first packet of a sequence.

EncryptedDataPacketHeader (xx byte) is defined in section 2.9.10.

EncryptedDataSequenceHeader (8 byte) the size in byte of the payload. It was prepended to the cleartext before the encryption. Used to know how much fragment to expect and to skip the last packet padding if present. Only in the first packet of a sequence.

285 **EncryptedPayload** data encrypted using AES256-GCM. The cleartext of the last packet in a sequence can be padded with cryptographically random bytes.

AD (16 byte) is an authentication tag appended to the ciphertext used by AES256-GCM encryption.

PacketTrailer (8 byte) is defined in section 2.9.3

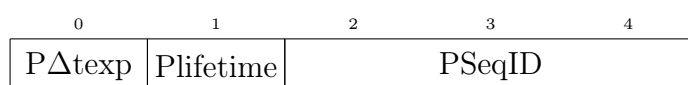
290 To calculate the size required for parts of a flowlet, subtract the size of headers and trailer from the given MTU. For the first packet of a sequence:

EncryptedPayload = Nexus MTU - (PacketHeader + SequenceHeader + EncryptedPacketHeader + EncryptedSequenceHeader + AD)

As sequence headers are present only in the first packet for successive packets the rule is:

295 EncryptedPayload = Nexus MTU - (PacketHeader + EncryptedPacketHeader + AD)

2.9.10 Encrypted Data Packet header



Nodes periodically update their symmetric shared secret keys, so that a compromise of a node keys will not decrypt old packets. Periodic rekey reduce the volume of data encrypted

300 under a single cipher key and reduce the time of its validity, a factor that can be a mitigation for possible future attacks.

A rekey MUST be done in any case before $2^{64} - 1$ packets of the same data plane sequence.

If a rekey failed for any reason the entry is removed from the peers table.

2.9.11 DATA-ACK Sequence

305 2.9.12 DATA-NACK Sequence

2.9.13 DATA-RENDEZVOUS Sequence

3 PROTOCOL EFFICIENCY

3.1 On Ethernet Nexus

The data plane protocol efficiency over a 1500 bytes MTU Ethernet Network is:

$$\begin{aligned} \text{Protocol Efficiency} &= \frac{\text{Payload Size}}{\text{Packet Size}} \\ \text{First packet} & \quad \frac{1500 - (8 + 70 + 8 + 8 + 16)}{1500} = 92.6\% \\ \text{Successive packets} & \quad \frac{1500 - (8 + 8 + 16)}{1500} = 97.8\% \end{aligned} \quad (2)$$

310 From 2 formulas when incoming data to be transmitted is much greater than Nexus MTU the protocol efficiency approach 97.8%. Instead if all the time, it is less than MTU the efficiency goes down to 92.6%.

Noise Protocol Framework nomenclature

315 NOVA uses Noise Protocol Framework[12] to perform the cryptographic operations among which P-Stratum connection handshake. In NOISE nomenclature, interactive patterns are named with two characters, which indicate the status of the initiator and responder's static keys.

The first character refers to the initiator's static key:

N = No static key for initiator

320 **K** = Static key for initiator Known to responder

X = Static key for initiator Xmitted ("transmitted") to responder

I = Static key for initiator Immediately transmitted to responder, despite reduced or absent identity hiding

The second character refers to the responder's static key:

325 **N** = No static key for responder

K = Static key for responder Known to initiator

X = Static key for responder Xmitted ("transmitted") to initiator

The token that will be exchanged in the following schemas are:

s : The local static key pair

330 **e** : The local ephemeral key pair

rs : The remote party's static public key

re : The remote party's ephemeral public key

ee, se, es, ss : A DH is performed between the initiator's key pair (determined by the first letter) and the responder's key pair (determined by the second letter).

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